

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify the dates and direction of each.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define the two ayanas.**
 - *Uttarāyaṇa* — northward course. Roughly Dec 21 (winter solstice) through Jun 21 (summer solstice). Sun moves NORTH on the sky (Northern Hemisphere).
 - *Dakṣiṇāyana* — southward course. Jun 21 through Dec 21. Sun moves SOUTH.
2. **Cultural importance.** *Uttarāyaṇa* is considered auspicious in many Indian traditions. *Makara Saṅkrānti* (~Jan 14) is celebrated as the start of *Uttarāyaṇa* — but technically the solstice (Dec 21) IS the start. The Jan 14 date reflects the sidereal-zodiac (*nirayana*) convention. We’ll unpack this on Day 7.
3. **Source.** Sūrya-siddhānta defines both ayanas in chapter 2.
4. **Modern bridge.** The Northern Hemisphere’s summer happens during the Sun’s *Uttarāyaṇa* (high northern declination). NCERT Geography IX ch. 3 has the standard diagram.
5. **Identify today.** “*Which ayana are we in today?*” Students compute based on the date. (Northern winter to summer = *Uttarāyaṇa*; rest = *Dakṣiṇāyana*.)

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Uttarāyaṇa and Dakṣiṇāyaṇa.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).